

LINEAR EQUATIONS

Top 2 Test

$$1. \quad a) \quad x - 11 = 24$$

$$x = 35$$

$$b) \quad x + 3 = 10$$

$$x = 7$$

$$c) \quad \frac{y}{8} = -9$$

$$y = -72$$

$$d) \quad \frac{m}{2} - 8 = 16$$

$$\frac{m}{2} = 24$$

$$m = 48$$

$$e) \quad 4x - 12 = 0$$

$$4x = 12$$

$$x = 3$$

$$f) \quad 3(x + 2) = 5$$

$$3x + 6 = 5$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

$$g) \quad 5p - 7 = 4p + 8$$

$$5p - 4p = 8 + 7$$

$$p = 15$$

$$h) \quad \frac{8x - 5}{3} = -2$$

$$8x - 5 = -6$$

$$8x = -1$$

$$x = -\frac{1}{8}$$

$$i) \quad \frac{4x}{9} = 5 + x$$

$$4x = 45 + 9x$$

$$4x - 9x = 45$$

$$-5x = 45$$

$$x = -9$$

$$j) \quad 20 = \frac{3x - 7}{4}$$

$$3x - 7 = 80$$

$$3x = 87$$

$$x = 29$$

$$k) \quad 5(3x - 2) = 3(2x - 10)$$

$$15x - 10 = 6x - 30$$

$$15x - 6x = -30 + 10$$

$$9x = -20$$

$$x = -\frac{20}{9}$$

$$l) \quad 4m - 3(m + 2) = 9$$

$$4m - 3m - 6 = 9$$

$$m = 9 + 6$$

$$m = 15$$

$$2. \quad \text{Let the number} = x$$

$$4x + 15 = x + 45$$

$$3x = 30$$

$$x = 10$$

$$3. a) \quad S = \frac{20}{2}(6 + 240)$$

$$= 2460$$

$$b) \quad 680 = \frac{n}{2}(5 + 75)$$

$$1360 = 80n$$

$$n = 17$$